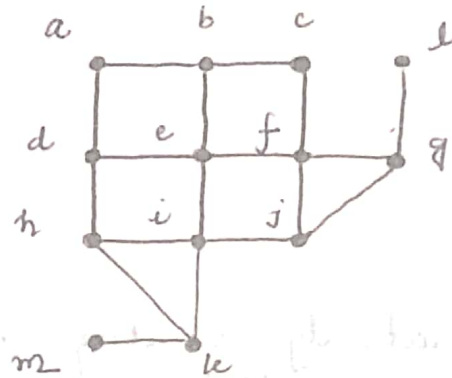


Previous Question Solve

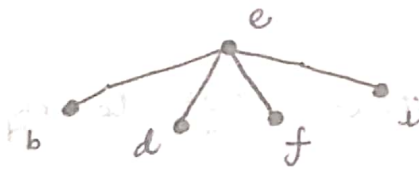
Topic : TREES.

Spring 2019.
Question 6.

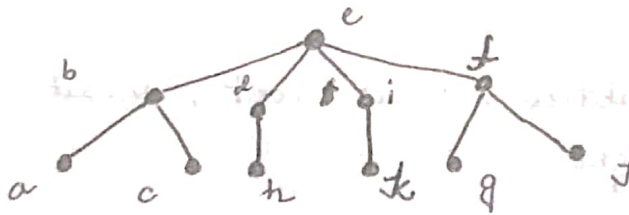
a) Use BFS to find a spanning tree for the following graph.



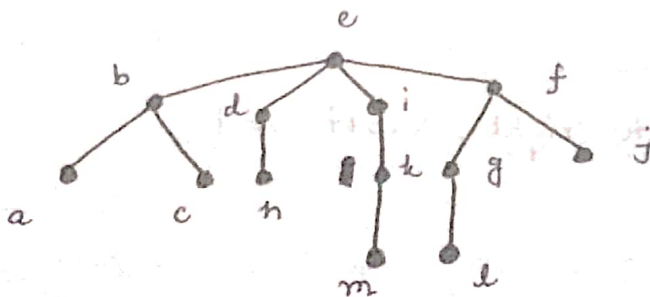
Solution :-



(1)



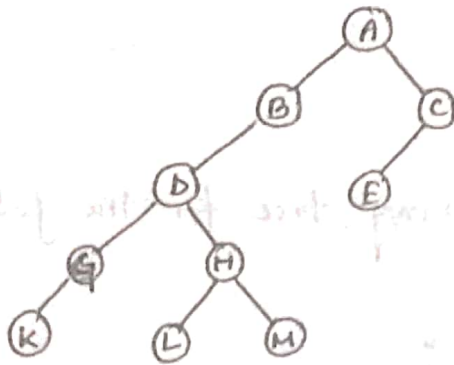
(2)



(3)

Answer

b1 what do you mean by tree traversing?
find the inorder, preorder and postorder traversal of the following tree.



Solution :-

Tree traversing :-

Procedures for systematically visiting every vertex of an ordered tree is called traversal.

Preorder traversal :-

Visit root, visit subtrees left to right.

A B D G K H L M C E

Inorder traversal :-

Visit left most subtree, visit root, visit other subtrees left to right.

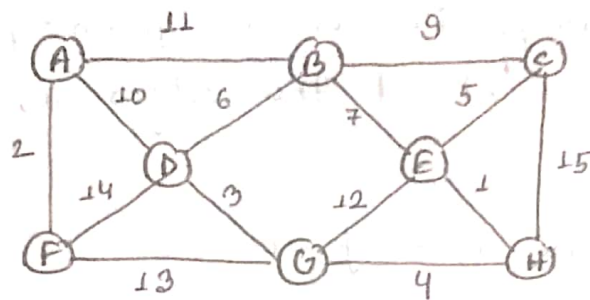
K G D L H M B A E C

Post order Traversal :-

Visit subtrees left to right, visit root.

K G L M H D B E C A

c) Define minimum spanning tree. Construct a minimum spanning tree of the following graph.

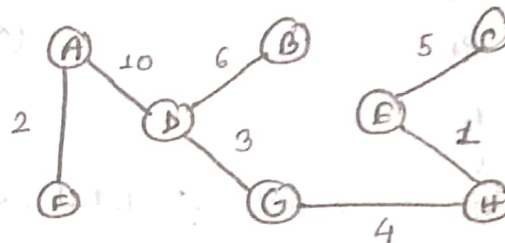


Solution :-

List of the edges in order of size

EH	1	BC	9
AF	2	AD	10
DG	3	AB	11
GH	4	EG	12
CE	5	FG	13
BD	6	DF	14
BE	7	CH	15

∴ The minimum spanning tree (Kruskal's method)



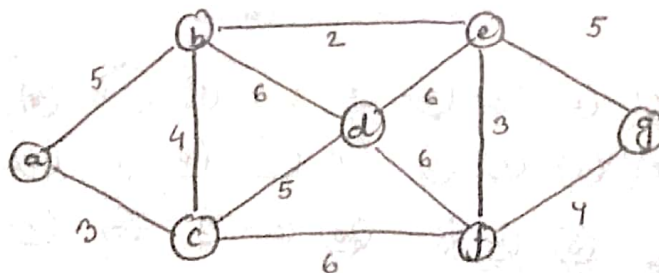
Total ~~tree~~ weight of tree =

$$1 + 2 + 3 + 4 + 5 + 6 + 10 = 31.$$

Autumn 2018.

Question 6.

a) Define minimum spanning tree. Construct minimum spanning tree of the following graph.



Solution:-

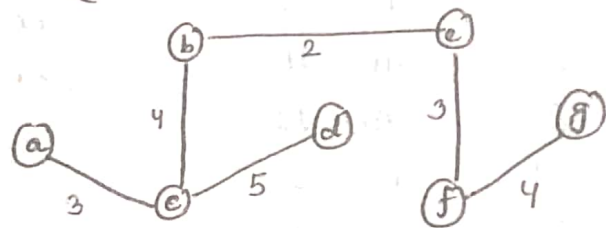
Minimum Spanning tree :-

A minimum spanning tree is a spanning tree which the sum of the weight of the edges is as minimum as possible.

List of the edges in order of size.

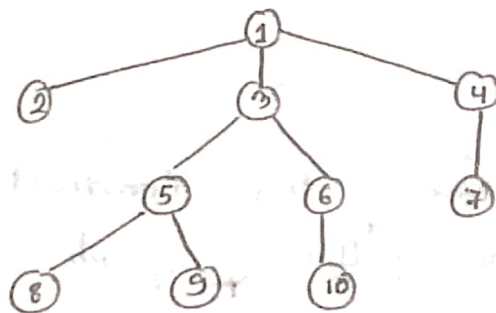
be	2	bd	6
ac	3	de	6
ef	3	df	6
bc	4		
fg	4		
ab	5		
cd	5		
eg	5		

∴ The minimum spanning tree:
(Kruskal's method)



$$\therefore \text{Total weight of tree} = 2 + 3 + 3 + 4 + 4 + 5 = 21.$$

b) Find the inorder, preorder, postorder traversal of the following tree.



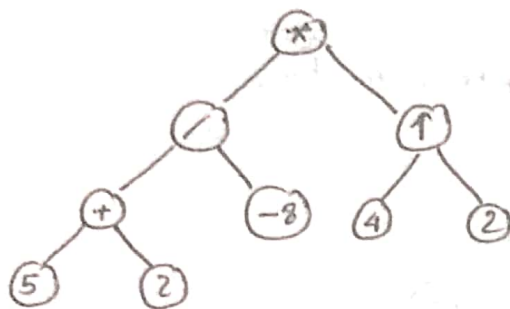
Solution :-

Pre order :- 1 2 8 5 8 9 6 10 4 7

In order :- 2 1 8 5 8 9 10 6 7 4

Post order :- 2 8 9 5 10 6 3 7 4 1

c) ① Binary tree for the expression,
 $((5+2) / -8) * (4^2)$



② Postfix notation :- $5\ 2\ +\ -8\ /\ 4\ 2\ \wedge\ *$

Spring 2018

Question 6.

Q1 what is m-ary tree? How many edges does a full binary tree with 100 internal vertices have?

Solution :-

m-ary tree :- A rooted tree is called m-ary tree if every internal vertex has no more than m children. It will be a full m-ary tree if every internal vertex has exactly m children.

b)

① Given, $E = (2x+y)(5a-b)^3$

$$= ((2x+y) * (5a-b) \uparrow 3)$$

tree T corresponds to expression E ,



② preorder traversal :- $(*) \rightarrow (+) \rightarrow (*) \rightarrow (2) \rightarrow (x) \rightarrow (y) \rightarrow (\uparrow)$

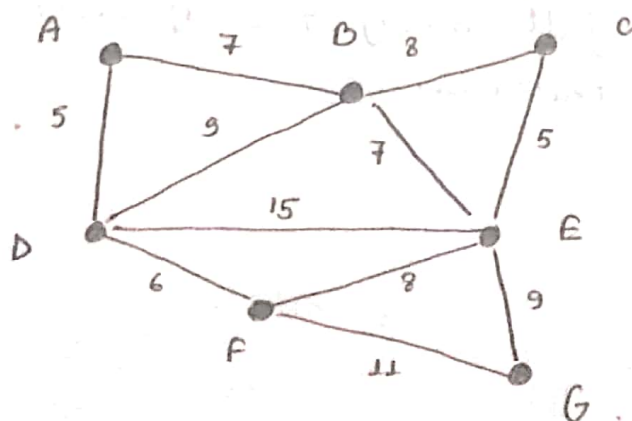
$\rightarrow (-) \rightarrow (*) \rightarrow (5) \rightarrow (a) \rightarrow (b) \rightarrow (3)$

postorder traversal :- $(2) \rightarrow (x) \rightarrow (*) \rightarrow (y) \rightarrow (+) \rightarrow (5) \rightarrow$

$\rightarrow (a) \rightarrow (*) \rightarrow (b) \rightarrow (-) \rightarrow (3) \rightarrow (\uparrow) \rightarrow (*)$

c)

Construct a minimum spanning tree.



list of all edges in minimum order.

$$AD = 5$$

$$EG = 9$$

$$CE = 5$$

$$FG = 11$$

$$DF = 6$$

$$DE = 15$$

$$AB = 7$$

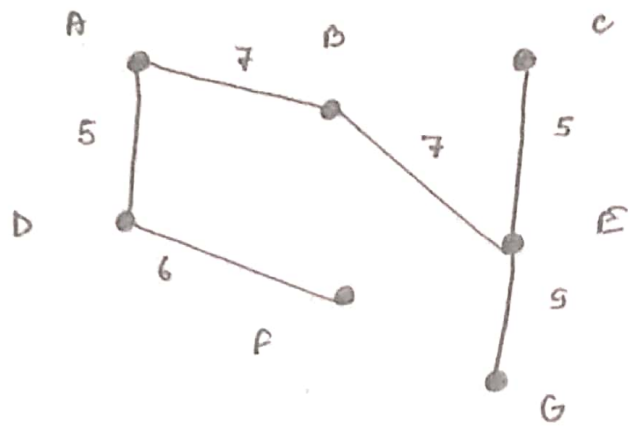
$$BE = 7$$

$$BC = 8$$

$$EF = 8$$

$$BD = 9$$

∴ The minimum spanning tree :-



$$\begin{aligned}\therefore \text{Total weight of the tree} &= 5 + 5 + 7 + 7 + 6 + 9 \\ &= 39\end{aligned}$$